# Requirements & Tech Stack

## Overview

This document captures the full functional requirements for the Animal Adoption Management System, organized by user role. Requirements are tagged with a priority (Must Have / Should Have / Nice to Have) and a scope flag (In Scope / Future Scope / Out of Scope) based on the initial requirements gathering session. It also defines the tech stack for the development of the system.

## 1. Functional Requirements

### 1.1 Adopter

| ID | Requirement | Priority | Scope |
|---|---|---|---|
| **A-01** | Users can create and update their profile with personal information including name, email, phone, address, age, and profile photo, identity & verification (Government ID type + number (for identity verification before adoption approval), ID verification status), household information (Housing type (apartment, house, condo), whether they own or rent, landlord contact / pet policy confirmation (critical — many adoptions fall through here), number of people in household, number and ages of children in household, lifestyle & compatibility (Employment status / work-from-home status, Activity level (sedentary, moderate, active), Yard availability (yes/no), Experience with pets (first-time owner, experienced, professional – pre-qualify), Current or past pets (species, breed, status), Adoption Preferences (Preferred species / breed, Preferred age range of pet, Preferred size, Open to special needs animals (yes/no)), Account & System Fields (Account creation date, Last login date, Account status (active, suspended, banned), Email verification status), Approval status, Adopter type - Fosterer/Owner | Must Have | In Scope |
| **A-02** | Users can search for animals using filters: species, breed, age, location, and compatibility with children or other pets. | Must Have | In Scope |
| **A-03** | Search results are scoped across all branches of the organisation (multi-shelter network). | Must Have | In Scope |
| **A-04** | Users can save animals to a favorites list and remove them at any time. | Must Have | In Scope |
| **A-05** | Users can submit an adoption application for a specific animal. | Must Have | In Scope |

| ID | Requirement | Priority | Scope |
|---|---|---|---|
| **A-06** | Users can track the status of their application (Pending, Approved, Declined, Withdraw), and surrender an approved pet back to the shelter; staff can revoke an approval, e.g. when the pet is never collected. | Must Have | In Scope |
| **A-07** | Users can schedule a shelter visit or virtual meet-and-greet with an animal. | Must Have | In Scope |
| **A-08** | Users can access pet care and training resource guides. | Should Have | Out of Scope |
| **A-09** | Automated reminders for vaccinations, vet visits, and grooming tasks post-adoption, where they want to be notified (email/SMS). | Must Have | In Scope |
| **A-10** | Users can share adoption success stories and photos. | Nice to Have | Out of Scope |
| **A-11** | AI adopter behavior rater – Based on adopter history, flags accounts who regularly return pets or don't attend vet appointments | Nice to Have | Out of Scope |
| **A-12** | Adopter History | Must Have | In Scope |
| **A-13** | Submitting an adoption application requires a non-refundable processing fee, paid via Stripe Checkout before the application record is created. | Must Have | In Scope |
| **A-14** | AI compatibility matcher: adopters take a short personality quiz and see their top matching available pets, each with a short 'why this match' reason. | Must Have | In Scope |

## 1.2 Shelter Staff

| ID | Requirement | Priority | Scope |
|---|---|---|---|
| **S-01** | Staff can approve and deactivate staff profile | Must Have | Must Have |
| **S-02** | Staff can add, update, and manage animal profiles including photos, species, breed, age, weight, height, blood group, behavior/personality, and colour, Microchip number (unique identifier, industry standard), Intake date, intake type (stray, surrendered, transferred from another shelter), adoption status (incoming, available, pending, adopted, fostered, transferred, deceased), shelterID | Must Have | In Scope |
| **S-03** | If a shelter is at full capacity, staff must assign the incoming animal to another shelter branch. | Must Have | In Scope |
| **S-04** | Staff can review, approve, or decline adoption applications. | Must Have | In Scope |
| **S-05** | Staff can communicate with applicants directly through the system. | Should Have | In Scope |
| **S-06** | Staff can manage volunteer applications, schedules, and task assignments. | Must Have | In Scope |

| ID | Requirement | Priority | Scope |
|---|---|---|---|
| **S-07** | Staff can accept and manage donations made to their shelter branch. | Must Have | In Scope |
| **S-08** | Staff can create and manage adoption events, fundraisers, and community programs. | Must Have | In Scope |
| **S-09** | Staff can track medical history, vaccinations, treatments, and special needs for each animal. | Must Have | In Scope |
| **S-10** | Staff can view reports on adoption rates, animal intake/outflow, and medical treatments. | Should Have | In Scope |
| **S-11** | Animal Transfer Management (Initiating a transfer request, Accepting/rejecting a transfer from another branch, Transfer history log) | Must Have | Must Have |
| **S-12** | Branch-scoped access: All staff actions are scoped to their assigned shelter branch by default. Cross-branch visibility is read-only (e.g. for transfer purposes). | Must Have | Must Have |
| **S-13** | Foster management | Nice to Have | Out of Scope |
| **S-14** | AI health passport summary: staff can generate a short summary of the pet's records plus points to review, each citing the record it came from; no diagnoses, medications or dosages. | Should Have | In Scope |

## 1.3 Veterinarian

| ID | Requirement | Priority | Scope |
|---|---|---|---|
| **V-01** | Vets can view and manage upcoming animal health appointments. | Must Have | In Scope |
| **V-02** | Vets can record vaccinations including vaccine name, type, status, last administered date, and due date. | Must Have | In Scope |
| **V-03** | Vets can access the universal health passport for any animal, even if it has transferred between shelter branches. | Must Have | In Scope |
| **V-04** | AI health passport summary: vets can generate a short summary of the pet's records plus points to review, each citing the record it came from; no diagnoses, medications or dosages. | Should Have | In Scope |

## 1.4 Volunteer

| ID | Requirement | Priority | Scope |
|---|---|---|---|
| **VL-01** | Volunteers can view their assigned tasks and current task status. | Must Have | In Scope |
| **VL-02** | Volunteers can view the status of their application (Pending, Approved, Rejected). | Must Have | In Scope |

| ID | Requirement | Priority | Scope |
|---|---|---|---|
| VL-03 | Volunteer Sign up – Basic information (name, email, phone, address, age, and profile photo, identity & verification (Government ID type + number (for identity verification before adoption approval), ID verification status)), task preferences, available timings, shelter ID | Must Have | In Scope |
| VL-04 | Volunteers can manage and update their schedule. | Must Have | In Scope |
| VL-05 | Volunteers can mark tasks as complete. | Must Have | In Scope |

## 1.5 Donor

| ID | Requirement | Priority | Scope |
|---|---|---|---|
| D-01 | Donors can make a monetary donation to a specific shelter branch. (Payment gateway) | Must Have | In Scope |
| D-02 | Donation receipt/confirmation (email) | Must Have | In Scope |
| D-02 | Sign up - Basic information (name, email, phone, address, age, and profile photo | Must Have | In Scope |
| D-03 | Donors can view their full donation history including date, amount, about the impact (TBD) and shelter. | Must Have | In Scope |

## 1.6 Admin

| ID | Requirement | Priority | Scope |
|---|---|---|---|
| AD-01 | Admin can create and manage shelter branch records including address and capacity. | Must Have | In Scope |
| AD-02 | Admin can assign a staff member as the designated manager of a shelter branch. | Must Have | In Scope |
| AD-03 | Admin has organisation-wide visibility into all shelter operations and transfers. | Must Have | In Scope |
| AD-04 | Admin can view system-wide reports/analytics. | Must Have | In Scope |

# 2. Non-Functional Requirements

| ID | Category | Requirement |
|---|---|---|
| NF-01 | **Security** | All passwords must be hashed using bcrypt. Authentication handled via JWT tokens with appropriate expiry. |
| NF-02 | **Security** | Role-based access control (RBAC) must restrict all API endpoints to authorised roles only. |
| NF-03 | **Performance** | API response times should remain under 500ms for standard CRUD operations under normal load. |

| ID | Category | Requirement |
|---|---|---|
| NF-04 | Scalability | The system must support multiple shelter branches without architectural changes — new branches added via admin panel only. |
| NF-05 | Availability | The application should target 99.5% uptime in production. Availability is monitored by UptimeRobot, which checks the API every 10 minutes and emails an alert on downtime. |
| NF-06 | Data Integrity | All adoption-critical data (applications, medical records, transfers) must be persisted with ACID-compliant transactions. |
| NF-07 | Geolocation | Location-based animal search must use PostGIS for efficient proximity queries across shelter branches. |
| NF-08 | Maintainability | Codebase must include unit tests (Jest), integration tests (Supertest), and end-to-end tests (Cypress) with minimum 70% coverage. |
| NF-09 | Usability | The UI must be responsive and accessible on both desktop and mobile browsers. |
| NF-10 | Data Privacy | Sensitive fields (government ID, passwords, medical records) must never be exposed in API responses beyond what the requesting role is authorised to see. |
| NF-11 | API Documentation | All endpoints must be documented via Swagger/OpenAPI. |
| NF-12 | Error Handling | The API must return consistent, structured error responses. |
| NF-13 | AI Usage | AI calls send no personal data (no names, contact details or addresses). AI is optional: the application works without it and falls back to rule-based scoring. |

# 3. Tech Stack

The following table outlines the tools and technologies chosen for the system, organized by architectural layer, along with the rationale behind each selection.

| Layer | Tool / Technology | Role | Why This Choice |
|---|---|---|---|
| **Frontend** | **React** | UI Component Library | Component-based architecture enables reusable UI elements across shelter portals and adopter dashboards; large ecosystem supports rapid feature development. |
| **Frontend** | **Tailwind CSS** | Utility-First Styling | Eliminates custom CSS overhead; consistent design system with responsive utilities ideal for a multi-role, multi-screen application. |
| **Frontend** | **React Router** | Client-Side Routing | Declarative routing for SPA navigation across role-specific views (Admin, Staff, Adopter, Vet) without full-page reloads. |
| **State Management** | **Zustand** | Global State | Lightweight and boilerplate-free compared to Redux; ideal for |

| Layer | Tool / Technology | Role | Why This Choice |
|---|---|---|---|
| | | | managing auth sessions and cross-component state like favorites and application status. |
| **State Management** | **TanStack Query** | Server State & Caching | Handles async data fetching, caching, and background sync - critical for real-time adoption status and appointment updates. |
| **Backend** | **Node.js + Express** | REST API Server | Non-blocking I/O suits concurrent shelter requests; Express provides a minimal, flexible framework for building the API layer quickly. |
| **Backend** | **multer** | File uploads | Parses multipart photo/ID uploads with type and 5 MB size limits. |
| **Backend** | **helmet** | HTTP security headers | Sensible secure defaults for every API response. |
| **Backend** | **zipcodes** | Offline geocoding | Turns US ZIP codes into coordinates for PostGIS without an external API. |
| **Backend** | **node-cron** | Scheduled jobs | Runs nightly maintenance jobs. |
| **ORM** | **Prisma** | Database Access Layer | Type-safe queries with auto-generated client reduce runtime errors; schema migrations are version-controlled and easy to maintain. |
| **Database** | **PostgreSQL** | Primary Relational DB | ACID compliance and strong relational support handle complex joins across pets, applications, and multi-shelter transfers reliably. |
| **Database** | **PostGIS** | Geospatial Extension | Enables location-based animal search and shelter proximity filtering - a core feature for the multi-shelter network. |
| **Database** | **Supabase** | DB Host + Media Storage | Managed Postgres hosting with built-in storage for pet photos and documents; reduces DevOps overhead. |
| **Storage** | **Supabase Storage** | File storage | Public pet-images and private government-ids buckets; private files served only via short-lived signed URLs. |
| **Auth** | **JWT** | Stateless Authentication | Scalable token-based auth supports multiple roles (Admin, Staff, Adopter, Vet) without server-side session storage. |
| **Auth** | **bcrypt** | Password Hashing | Industry-standard adaptive hashing algorithm protects user credentials |

| Layer | Tool / Technology | Role | Why This Choice |
|---|---|---|---|
| | | | with salted hashes resistant to brute-force attacks. |
| **Containers** | **Docker** | Containerization | Ensures environment parity across development, testing, and production; simplifies onboarding for new contributors. |
| **Version Control** | **GitHub** | Source Control | Industry-standard Git hosting with PR workflows, branch protection, and CI/CD integration. |
| **API Docs** | **Swagger/OpenAPI** | API Documentation | Auto-generates interactive API docs from annotations - facilitates frontend-backend handoffs and third-party integrations. |
| **Payments** | **Stripe** | Payment Processing (Checkout) | Handles the $15 adoption-application fee end-to-end via hosted Checkout + webhook-confirmed payment — card data never touches the app, so PCI scope stays entirely with Stripe. |
| **AI** | **Groq (openai/gpt-oss-20b)** | Matcher reasons + passport summary | Free tier; open-weight model; OpenAI-compatible API called with plain fetch from the server (no SDK), so switching provider (e.g. local Ollama) is an env change. |
| **Testing** | **Supertest** | HTTP Assertions for integration tests | Drives the real Express app over HTTP inside Jest (register → login → call the actual endpoint), so integration tests exercise the full request/response cycle, not isolated functions. |
| **Testing** | **Cypress** | End-to-end browser tests | Drives the real app in an actual browser for flows that only exist client-side (e.g. the post-login redirect-back flow) — Jest/Supertest never render React, so they can't reach these. |
| **Testing** | **Jest** | Unit & Integration Tests | Fast, zero-config test runner for backend logic and utility functions; integrates cleanly with Node/Express. |
| **Deployment** | **Render** | Backend Deployment | Managed cloud platform that runs the API's Docker image on a free plan with no expiry (the database stays on Supabase) and redeploys from GitHub, with minimal DevOps overhead. |
| **Development** | **Vercel** | Frontend Deployment | Zero-configuration deployment for React apps. Connects directly to |

| Layer | Tool / Technology | Role | Why This Choice |
|---|---|---|---|
| | | | GitHub and auto-deploys on every push, with a global CDN for fast load times out of the box. |
| **Monitoring** | **UptimeRobot** | Uptime Monitoring | Free monitor that calls the API every 10 minutes, so Render's free plan never sleeps (no 30–60 s cold start) and the Supabase database stays active; emails an alert if the API goes down. |
| **Development** | **Cursor** | AI-Powered IDE | AI-powered IDE that understands the entire codebase as context, enabling multi-file edits across React, Express, and Prisma layers in a single instruction. Ideal for a full-stack project with multiple interconnected portals. |
| **Planning** | **Trello** | Task & Sprint Board | Visual Kanban-style board for tracking sprint tasks, backlog grooming, and progress across the 8-sprint plan; simple and effective for solo project management. |
| **Design** | **Figma** | UI Prototyping | Collaborative design tool for wireframing and high-fidelity prototypes before frontend implementation. |
| **Design** | **dbdiagram.io** | ER-Diagram | Dedicated ER diagram creation tool. |
| **Planning** | **UXPressia** | User Persona Creation | Dedicated persona creation tool that structures user research into detailed, visually rich profiles — ideal for defining the goals, pain points, and behavioural patterns of the six user types (Admin, Shelter Staff, Adopter, Vet, Volunteer and Donor) before frontend design begins. |
| **Docs** | **Word/Google Docs** | Project Documentation | Centralized knowledge base for requirements, sprint planning, API references and development documentation. |